Practice Problems on Combinatorics

1. A family of eight people want to go to the airport. They call two taxi cabs. In how many different ways can they get on the cabs if:

   Each taxicab has room for a maximum of five passengers.

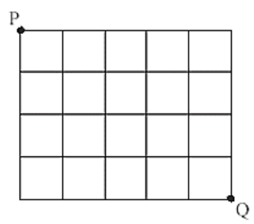
   Which chair a person sits on or next to whom they sit does not make a difference.

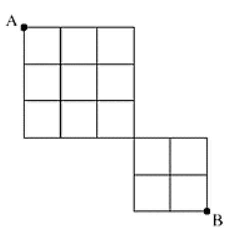
   Which cab a person rides on does not make a difference.

   Which group of people a person rides with is important and we would like to count the different possible combinations of that.

Note that we are looking for a solutions that satisfies all the above requirements.

2. Moving only right or down, how many paths are there from P to Q in the first grid? How many paths from A to B in the second grid?





3. In what percent of 12-digit strings that can be made with 4A's, 4B's, and 4C's all of the A's are next to each other?

4. How many 5-digit integers have no 9's ? Note that 00087 is not a 5-digit integer (it is a 2-digit integer), but 80770 is a 5-digit integer.

5. In a class of 30 students, how many committees of 3 people can be selected if the roles they are selected for is the same?

What if the roles are different?

6. When you play lotto 6-49, you choose 6 different integers from 1 to 49 inclusive. The order in which the numbers are chosen is not important. How many combinations are possible?

If the order at which the number were picked would matter, how many choices would be possible?

7. Assume a car license plate consists of 7 characters. The first 3 can be A to F (inclusive, uppercase) but no letter can be repeated. The next 3 characters can be any of the digits from 1 to 9 (inclusive) with possible repetition. The last character can be any of the letters X, Y, Z.

How many license plates are possible?

8. What percent of different arrangements of all the letters of the word OOMPALOOMPA have all the O's together? Example: MPALOOOOMPA

9. An IPv4- class B address, used for large network consists of 32 binary bits. It starts with 10 followed by a 14-bit network number, followed by a 16-bit local address. Local addresses of all 0's or all 1's are not allowed. There are no restrictions on the network number. How many different IPv4-class B addresses exist?

10. How many hexadecimal strings of length 12:
    a) Have no 9's and have at least one F?

    b) Start with 1A1A and contain 9EF (in this order with nothing between 9 and E and nothing between E and F) somewhere in the string? Example: **1A1A**007**9EF**3A

11. With A, B, C, M, N, P , 3, 4, 5, 6, 7:

    How many passwords of length 7 can we make to meet all of these requirements:

    Repetition of letters is allowed, but repetition of digits is not.

    The password must start with a letter and must end in a digit

    The password must have exactly 4 letters and exactly 3 digits.

    An example: AA76BB3

12. In a film festival ten short movies are to be played in sequence (one after another).

We are tasked with arranging the order in which these short movies are to be played on the screen.

Two of the shorts are in French, the rest are in English.

Three of the English shorts are about global warming.

We are asked to arrange the play sequence so that:

The first and last movies are about global warming, and the two French movies are not placed back-to-back (there is at least one English movie between the French ones).

How many arrangements meet the requirements?

13. How many ways are there to choose 10 co-op candidates out of 22 students to send to work at the same company if four of the students insist on being together (you can select all four of them or none of them) and another two students won't work together (you can select none of them or either one of them, but not both of them)? Hint: make a case-by-case argument, for example, one case is: employ all 4 friends and one of the enemies and 5 regulars …

14. Consider the equation $x_1 + x_2 + x_3 + x_4 = 17$
    A. How many positive <u>integer</u> solutions are there if the only restriction is that $x_i \geq 0$ Give **two different examples** of integer solutions for this equation.

    B. How many integer solutions of the previous equation exist if we require each $x_i \geq 2$?

15. How many arrangements of the word **MICROBLOGGING** have at least three letters between the O's and do not end with O?

16. What percent of different arrangements of all the letters of the word ELEMENT have all the E's together? Example: TEEENML

17. BCIT offers two coding bootcamps for the students who do not have coding experience. One is a Java bootcamp and the other is Python. Each bootcamp is booked in a room that has a max capacity of 6 people. Eight students are randomly assigned to the bootcamps in such a way that all students have a place to go, and no room will be over capacity. In how many different ways can people be assigned to these two bootcamps?
    Note that:
    The bootcamps run at the same time and we cannot enroll the same student in both.
    Java ABCD Python EFGH is a different case from Python ABCD JAVA EFGH
    Java ABCD Python EFGH is the same case as Java CBDA Python GHFE

18. A certain financial institution issues credit cards that are numbered as follows: a 6-digit IIN (issuer identification number) followed by a 16-digit PAN (primary account number).

IIN has to be either between 222100 to 272099 (inclusive) or, between 510000 to 559999 (inclusive).

The PAN cannot be all zeros.

  a) How many valid credit card numbers can be generated following the above requirements?
  b) How many valid credit card numbers in part a) have a PAN that start and end with 7 and do not contain the digit 5 (no extra restrictions on IIN other than what is given in the problem).

19. John Snow wants to give 13 Game Of Thrones shirts to six friends. The shirts are identical. In how many ways can he do that if:
  a) There are no restrictions
  b) John's favorite friend (Samwel Tarley) is to receive at least 5 shirts.

20. As the manager in charge of making the weekly shift schedule, you are assigning four employees to the following grid:

Let's call the four employees A, B, C, and D (short for Ali, Brad, Carmen, and David)

Each employee gets exactly four shifts in total (could be on the same day or different days).Here is an example of what the schedule could look like:

	Shift 1	Shift 2	Shift 3	Shift 4
Monday	B	B	D	A
Tuesday	A	D	A	D
Wednesday	D	B	C	C
Thursday	B	C	A	C

a) In how many ways can you assign the shifts?

b) Ali has asked you to give him the first shifts on Monday and Tuesday. Other than this, he is indifferent towards any other slots, so he is okay to get any of the remaining shifts as long as he has a total of 4 shifts as stated above. Everyone else needs 4 shifts and have made no special requests. In how many ways can you set the schedule?

21. A computer system considers a string of decimal digits to be a valid code word if and only if the string is 6 digits long, it starts with an even number, and does not end in an even number. Repetition is allowed.

    For example, **0**1222**7** is an acceptable sequence while 78658**8** is not an acceptable sequence.

    a) How many 6-digit strings are acceptable?

    b) In what percent of the acceptable strings, there are exactly four even digits?

22. Each of the commands in line 6 and line 9 take 2 units of time to execute (each time). The *if-command* in line 5 takes 7 units of time to execute (each time). All other lines take an insignificant amount of time. How much time is needed for the entire code block to execute?

```
1   Counter = 7
2   Counter = 12
3   for (i = 1 to 400) {
4       for(j = i to 400) {
5           if( i≠j){
6               counter=counter+6
7           }
8           for (k=12 to 153){
9               counter=counter+12
10          }
11      }
12  }
```

Number of times line 6 is executed:

Number of times line 5 is executed:

Number of times line 9 is executed:

Total amount of time for the entire code block:

23. Your friend has 10 coins in her pocket: Toonies, Loonies, Quarters and Dimes. If you correctly guess how many coins she has, you will win the coins.

a. What is the probability that you guess the number of each coin type correctly?

b. As a hint, she tells you that she has at least four of her coins are quarters and asks you to guess the number of other coins. Find the probability of correctly guessing this time.